Server Virtualization

Agenda

- Objectives
- Role of Server Virtualization
- Role of Hypervisor
- Types of Virtualization
- Managing Virtual Machines Resources
- Scaling up Virtual Machines
- Configuring Virtual Networking in Virtual Machines
- Summary

Objectives

- Describe the need for virtualization on servers
- Identify the usage of server virtualization.
- Identify 2 types of hypervisors
- Identify the 3 different types of virtualization
- Describe the types of advanced virtual machine operations used in server management.
- Understand how virtual machine scalability is achieved.
- Describe how networks in virtual servers are connected

Why the need for virtualization?

- Effective Resource Usage – Resources (RAM/CPU/DISK) can be allocated wherever needed.
- Ease of Management -
  - Unused machines can be kept indefinitely off until when required, and do not consume any resources other than disk space.
  - Application provisioning, maintenance, high availability and disaster recovery
- Security –
  - Applications can be kept on separate servers.
  - Separate servers do not pose a risk to each other
  - Especially true when virtual machines are owned by mutually untrusting users.

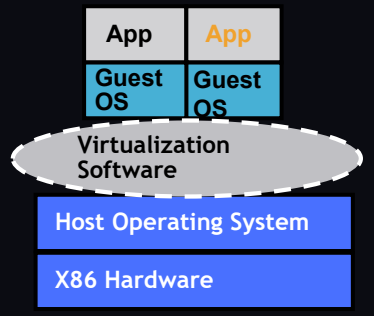
Server Virtualization Scenarios

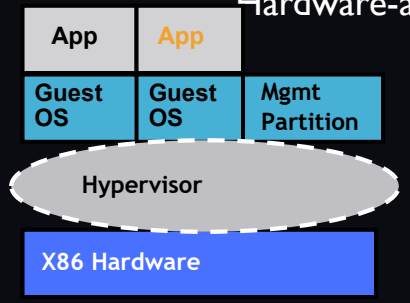
- Hardware-based virtualization
- Software-based virtualization
  - Hosted (application virtualization)
  - Hypervisor
  - Full virtualization (binary translation)
  - Para-virtualization (OS assisted)
  - Hardware-assisted virtualization (Intel VT-x/AMD-V)
- Diagram: Hosted (X86 Hardware > Host OS > Virtualization Software > Guest OS + App) vs. Hypervisor (X86 Hardware > Hypervisor > Mgmt Partition + Guest OS + App)

Hardware-based virtualization

Software-based virtualization

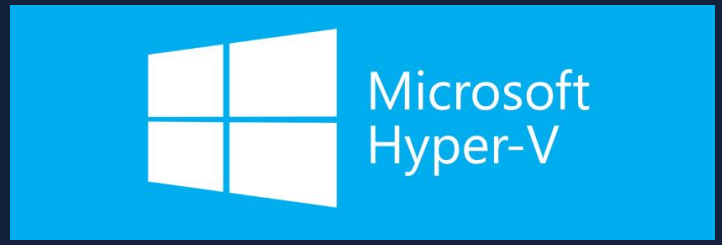
- Hosted (application virtualization)
- Hypervisor
  - Full virtualization (binary translation)
  - Para-virtualization (OS assisted)
  - Hardware-assisted virtualization (Intel VT-x/AMD-V)





Common Windows Hypervisors

- Microsoft Hyper-V
  - A native hypervisor included in Windows Server and Windows 10/11 Pro and Enterprise editions.
  - Supports both Type 1 (bare-metal) and Type 2 (hosted) virtualization.

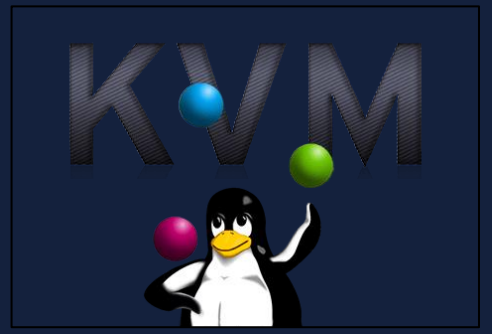


Common Linux Hypervisors

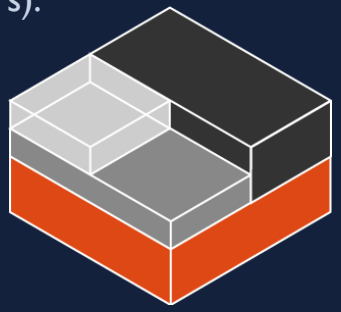
- Linux Open-Source Virtualization
- Xen (para-virtualized – same hardware as host)
  - A Type 1 hypervisor that allows multiple operating systems to run concurrently on a host machine.
  - Supports paravirtualization and hardware virtualization.



- KVM (Kernel Based Virtual Machine)-
  - A Type 1 hypervisor integrated into the Linux kernel.
  - Converts the Linux kernel into a bare-metal hypervisor.
  - Hardware Assisted -/ Emulated Hardware for VMs



- LXC (Linux Containers)
  - A lightweight virtualization method that uses the host's Linux kernel to run multiple isolated Linux systems (containers).
  - Not a traditional hypervisor but provides OS-level virtualization.



Commercial Hypervisors

- VMware vSphere/ESXi
  - A leading enterprise-level hypervisor for server virtualization.
  - Offers features like high availability, fault tolerance, and vMotion.

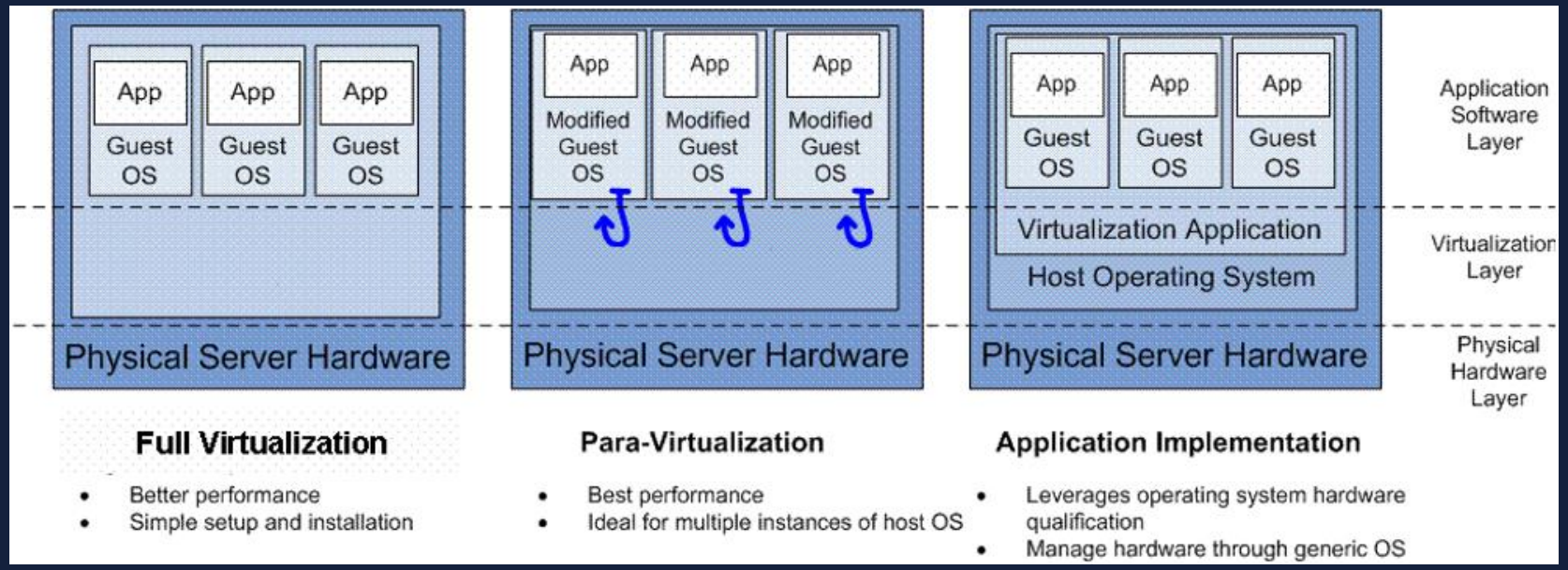


- Proxmox VE (Virtual Environment)
  - Proxmox VE is an open-source virtualization management platform that combines two virtualization technologies: KVM (Kernel-based Virtual Machine) for full virtualization and LXC (Linux Containers) for lightweight container-based virtualization.



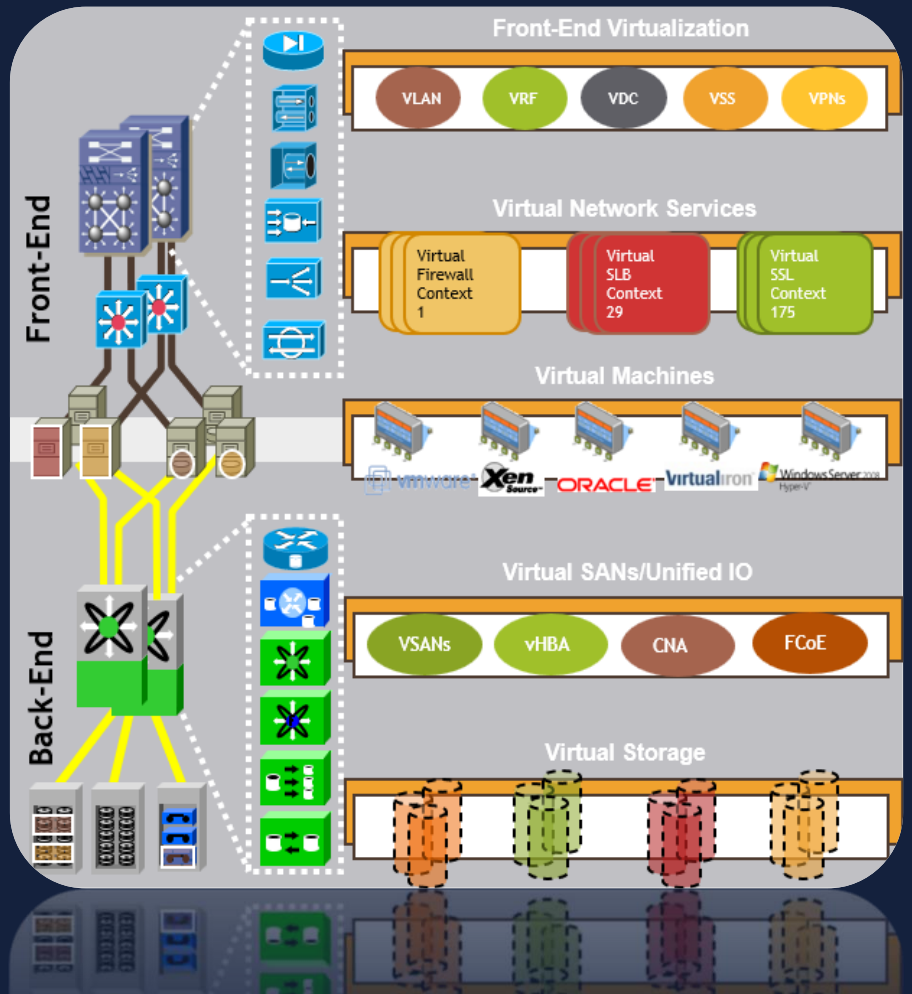
Software-Based Virtualization (Examples)

- Full Virtualization: better performance; simple setup and installation. Examples: VMware ESXi, Microsoft HyperV, Xen (with AMD-SVM or Intel VM-T), VirtualIron (hardware-assisted)
- Para-Virtualization: best performance; ideal for multiple instances of host OS. Examples: Xen (with traditional hardware), Oracle VM server
- Application Virtualization: leverages operating system hardware qualification; manages hardware through generic OS. Examples: Microsoft App-V, Citrix Virtual Apps



Hardware Assisted Virtualization

- Hardware-assisted virtualization is a virtualization approach that enables efficient full virtualization using help from hardware capabilities, primarily from the host processors.

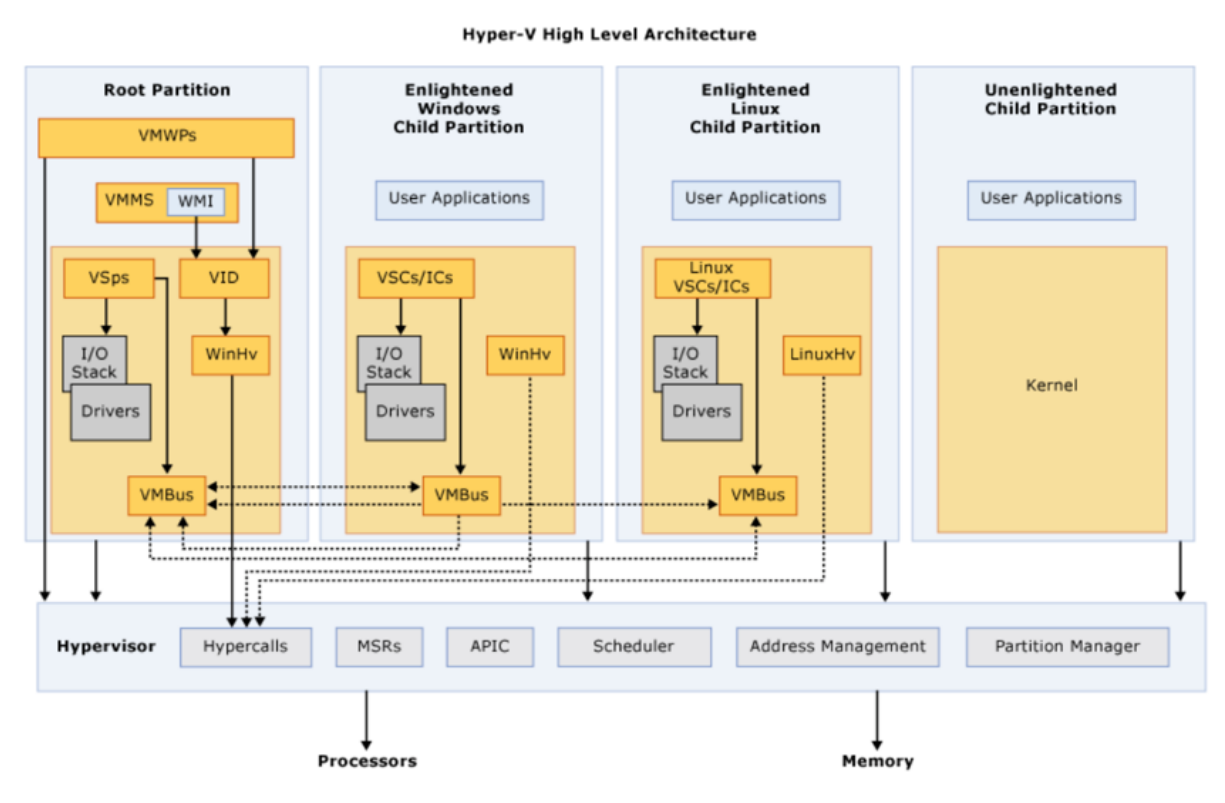


Hardware Assisted Virtualization

- Full virtualization on x86 has significant costs in hypervisor complexity and run-time performance.
- Same as Full Virtualization, however the Guest VM can communicate directly with the hardware bypassing the hypervisor for certain machine instructions.
- Explicit hardware support in the host CPU must be present.
  - Hardware-Assisted CPU Virtualization (Intel VT-x and AMD AMD-V)
  - Hardware-Assisted MMU Virtualization (Intel EPT and AMD RVI)
  - Hardware-Assisted I/O MMU Virtualization (Intel VT-d and AMD AMD-Vi)

Hardware Assisted Virtualization

- These extensions address the parts of x86 that are difficult or inefficient to virtualize, providing additional support to the hypervisor. This enables simpler virtualization code and a higher performance for full virtualization.
- Offers some speed improvements as instructions handled in hardware as compared to software (hypervisor)
- Diagram: Hyper-V High Level Architecture (Root Partition, Enlightened Windows/Linux Child Partitions, Unenlightened Child Partition, Hypervisor)



Enabling Hardware Assisted Virtualization - CPU

- First-generation hardware support for x86 virtualization with AMD-Virtualization (AMD-V) and Intel VT-x technologies in 2006
- Requires Hardware Data Execution Prevention: prevent an application or service from executing code from a non-executable memory region. This helps prevent certain exploits that store code via a buffer overflow, for example.
- Intel VT-x (initially codenamed Vanderpool)
  - Intel refers to it as Execute Disable (XD). This feature must be enabled in the system BIOS.
- AMD AMD-V (also called SVM and initially codename Pacifica)
  - AMD refers to it as No Execute (NX). This feature must be enabled in the system BIOS.
- When you enable hardware assisted virtualization (Intel VT or AMD AMD-V) in the BIOS, you must POWER CYCLE the system. NOT REBOOT.

Enabling Hardware Assisted Virtualization - MMU

- Intel introduced its second generation of hardware support that incorporates Memory Management Unit (MMU) virtualization, called Extended Page Tables (EPT). AMD also introduced its own MMU virtualization, called Rapid Virtualization Indexing (RVI) or Nested Page Tables (NPT)

- EPT/RVI-enabled systems can improve performance compared to using native shadow paging for MMU virtualization.
  - EPT provides performance gains of up to 48% for MMU-intensive benchmarks and up to 600% for MMU-intensive microbenchmarks
  - RVI provides performance gains of up to 42% for MMU-intensive benchmarks and up to 500% for MMU-intensive microbenchmarks

- VMM can now rely on hardware to eliminate the need for shadow page tables. This removes much of the overhead otherwise incurred to keep the shadow page tables up-to-date
- Turn these features on in BIOS to enable.

Enabling Hardware Assisted Virtualization – I/O

- A newer processor feature is an I/O memory management unit that remaps I/O DMA transfers and device interrupts. This can allow virtual machines to have direct access to hardware I/O devices, such as network cards.

- In AMD processors this feature is called AMD I/O Virtualization (AMD-Vi or IOMMU) and in Intel processors the feature is called Intel Virtualization Technology for Directed I/O (VT-d).

- In VMWare, VMDirectPath I/O leverages Intel VT-d and AMD-Vi hardware support to allow guest operating systems to directly access hardware devices. In the case of networking, VMDirectPath I/O allows the virtual machine to access a physical NIC directly rather than using an emulated or a para-virtualized device

- While VMDirectPath I/O has limited impact on throughput, it reduces CPU cost for networking-intensive workloads

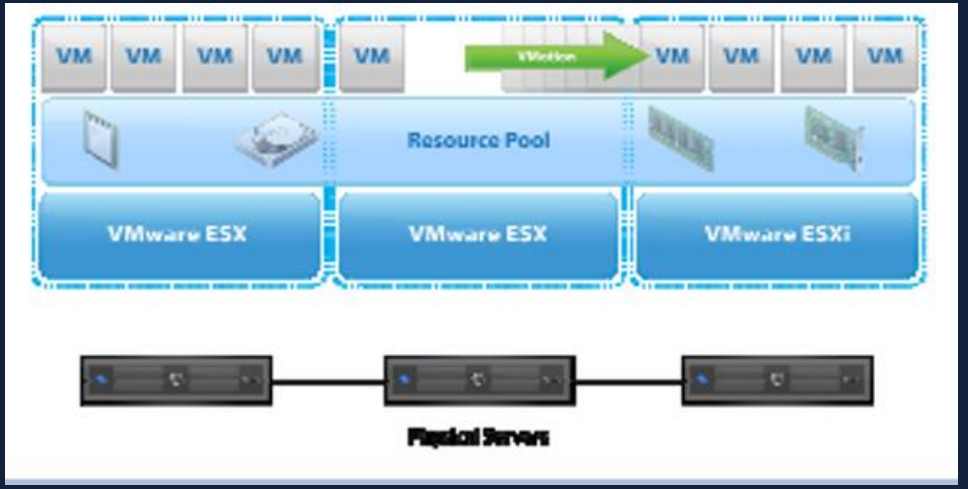
VMware Type 1 and Type 2 offerings

	Workstation Player	Workstation Pro	vSphere Essentials Plus Kit	vSphere Standard	vSphere Foundation
Hypervisor Type	Hosted	Hosted	Bare Metal	Bare Metal	Bare Metal
Typical Use Case	Test	Test & Dev	Data centers and cloud infrastructure	Data centers and cloud infrastructure	Data centers and cloud infrastructure
# of VMs per host	1-2	< 10	< 100	<100	<100
Dedicated Server Required	No	No	Yes	Yes	Yes
Centralized Management	No	No	Yes	Yes	Yes
Remote Management	No	No	Yes	Yes	Yes
Licensing Model	Free for personal use	Perpetual license	Subscription-Core based (Kit includes 96 core licenses)	Subscription-Per Core	Subscription-Per Core

Comparing ESXi to VMware vSphere

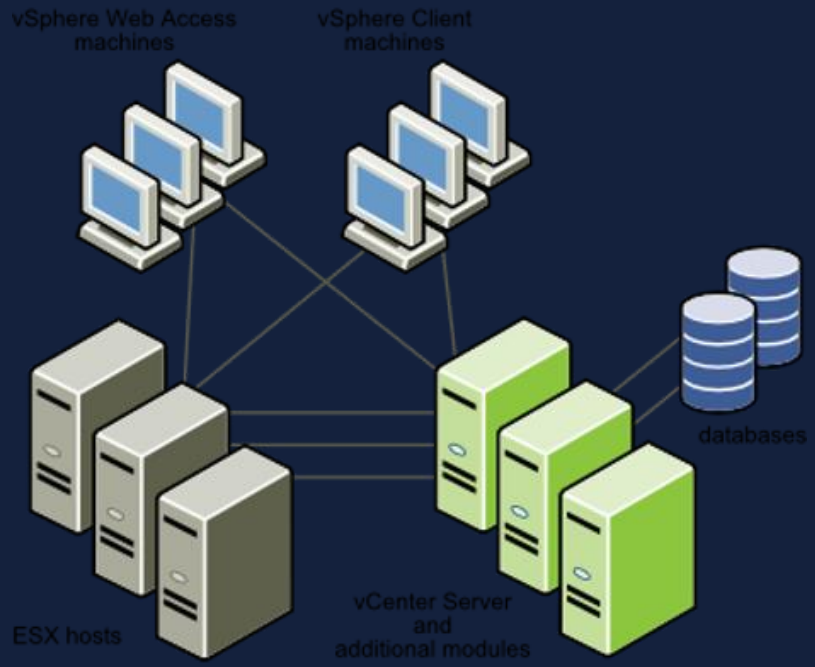
- VMware vSphere Hypervisor (ESXi)
  - Single server partitioning
  - Production-class hypervisor
  - Advanced server resource management
  - FREE
- VMware vSphere
  - Pools of computing resources
  - Centralized management
  - Built-in automation, availability and manageability
  - All Editions include ESXi





VMware vSphere Components

- Each vCenter Server system manages multiple ESXi hosts. You can run the vSphere Client and vSphere Web Access on multiple workstations.
- The major VMware vSphere components are:
  - VMware ESXi: Provides a virtualization layer that abstracts the processor, memory, storage, and networking resources of the physical host into multiple virtual machines.
  - vCenter Server: A service that acts as a central administration point for ESX/ESXi hosts connected on a network. This service directs actions on the virtual machines and the hosts. The vCenter Server is the working core of vCenter.



VMware vSphere Components

- vCenter Server additional modules: Provide additional capabilities eg. vCenter Update Manager, vCenter Converter, and vCenter Guided Consolidation Service and features to vCenter Server.

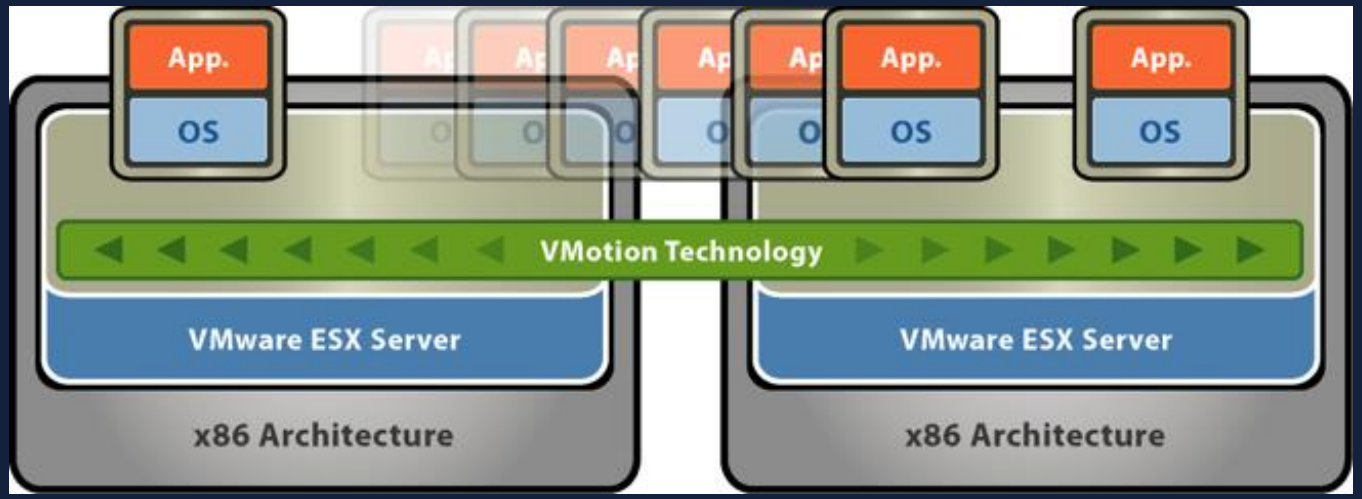
- vSphere Client: Installs on a Windows machine and is the primary method of interaction with VMware vSphere. The vSphere Client acts as a console to operate virtual machines and as an administration interface into the vCenter Server systems and ESX hosts.

- VMware vSphere Web Access: A browser-based interface for system administrators who need to access virtual machines remotely or without a vSphere Client.

- Databases: Organize all the configuration data for the VMware vSphere environment.

VMotion

- VMotion allows you to quickly move an entire running virtual machine from one host to another without any downtime or interruption to the virtual machine. This is also known as a "hot" or "live" migration.



VMotion

- The entire state of a virtual machine is encapsulated and the VMFS file system allows both the source and the target ESX host to access the virtual machine files concurrently. The active memory and precise execution state of a virtual machine can then be rapidly transmitted over a high speed network. The virtual machine retains its network identity and connections, ensuring a seamless migration process.

VMotion Operations

1) Migration request is made to move the virtual machine from ESX1 to ESX2.

2) vCenter Server verifies that the virtual machine is in a stable state on ESX1 and checks the compatibility of ESX2 (CPU, networking, etc.) to ensure that it matches that of ESX1.

3) The virtual machine is registered on ESX2.

4) The virtual machine state information (including memory, registers and network connections) is copied to ESX2. Additional changes are copied to a memory bitmap on ESX1.

5) The virtual machine is quiesced on ESX1 and the memory bitmap is copied to ESX2.

6) The virtual machine is started on ESX2 and all requests for the virtual machine are now directed to ESX2.

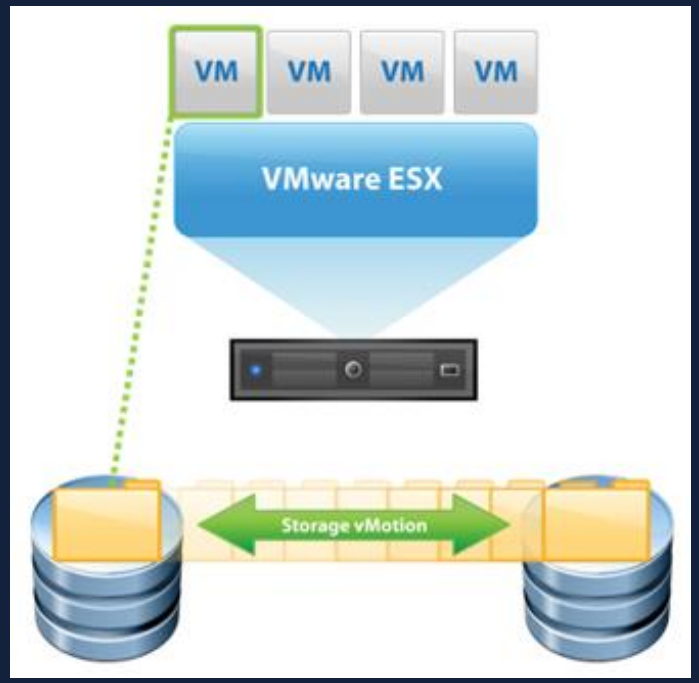
7) A final copy of the virtual machine's memory is done from ESX1 to ESX2.

8) The virtual machine is unregistered from ESX1.

9) The virtual machine resumes operation on ESX2.

Storage VMotion

- Storage VMotion was a new feature introduced in ESX 3.5, it allows you to migrate a running virtual machine and its disk files from one datastore to another on the same ESX host.
- VMotion simply moves a virtual machine from one ESX host to another but keeps the storage location of the VM the same
- Storage VMotion changes the storage location of the virtual machine while it is running and moves it to another datastore on the same ESX host. The virtual machine can be moved to any datastore on the ESX host which includes local and shared storage.



Storage VMotion Operations

1) New virtual machine directory is created on the target datastore, virtual machine configuration files and all non-virtual disk files are copied to the target directory.

2) ESX host does a "self" VMotion to the target directory.

3) A snapshot (without memory) is taken of the virtual machines disks in the source directory.

4) Virtual machine disk files are copied to the target directory.

5) Snapshot that is located in the source directory is consolidated into the virtual machine disk files located in the target directory.

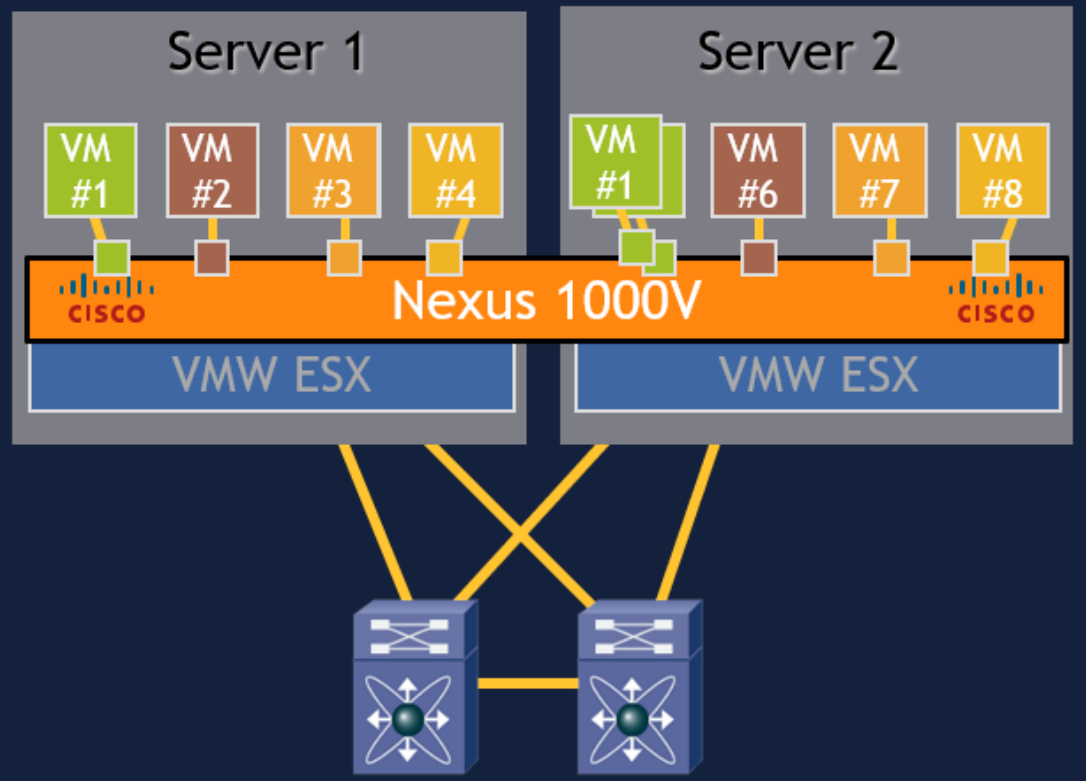
6) Source disk files and directory are deleted.

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- Objectives
- Role of Server Virtualization
- Role of Hypervisor
- Types of Virtualization
- Managing Virtual Machines Resources
- **Scaling up Virtual Machines**
- Configuring Virtual Networking in Virtual Machines
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Cisco Nexus 1000V

- Industry First 3rd Party Virtual Distributed Switch
- Nexus 1000V provides enhanced VM switching for VMW ESX environments
- Features VN-Link capabilities:
  - Policy-based VM connectivity
  - Mobility of network and security properties
  - Non-disruptive operational model
- Ensures visibility and continued connectivity during VMotion
- Enabling Acceleration of Server Virtualization Benefits



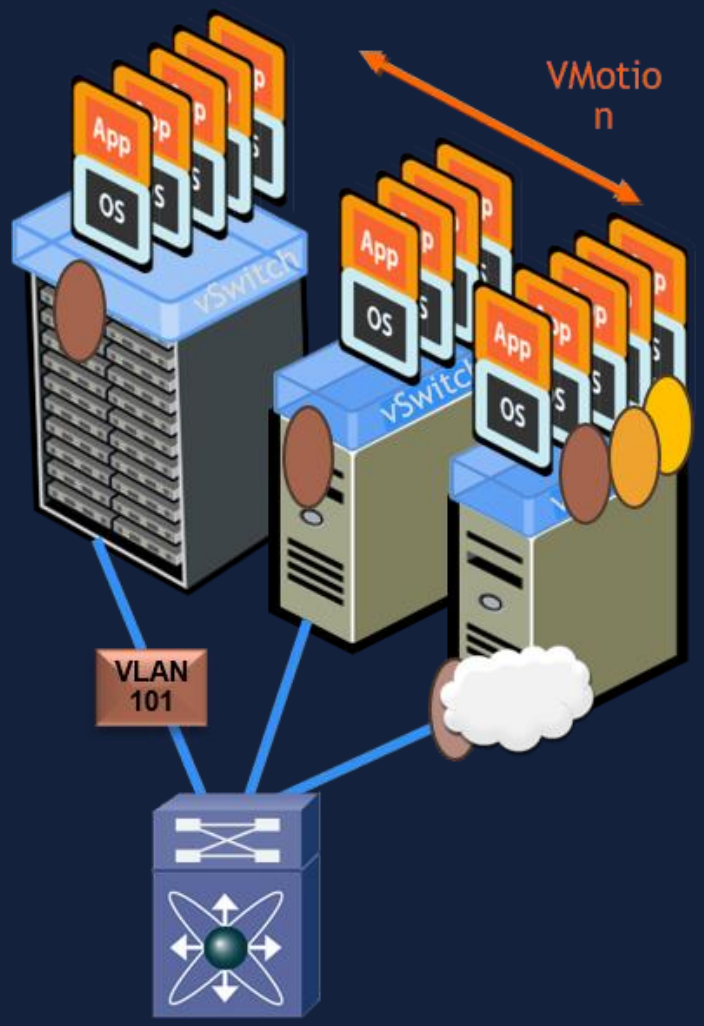
Enabling Acceleration of Server Virtualization Benefits

Newer versions after Cisco Nexus 1000v

- Cisco ACI (Application Centric Infrastructure)
  - Overview: ACI is a software-defined networking (SDN) solution designed to provide a more automated and application-centric approach to networking.
  - Features: It offers centralized management, policy-driven automation, and integration with various cloud environments.
- Cisco Nexus 9000 Series
  - Overview: This series includes both traditional and programmable switches that support a wide range of deployment models, including data centre, cloud, and enterprise.
  - Features: Supports Cisco's ACI and provides advanced features for high-performance networking.
- VMware NSX
  - Overview: A network virtualization platform that enables the creation of entire networks in software, allowing for flexibility and scalability in virtualized environments.
  - Features: Provides micro-segmentation, security, and integration with cloud services.
- Cisco Meraki
  - Overview: A cloud-managed networking solution that simplifies network management and provides visibility and control over network devices.
  - Features: Suitable for organizations looking for a simplified, cloud-based approach to networking.

VN-Link Brings VM Level Granularity

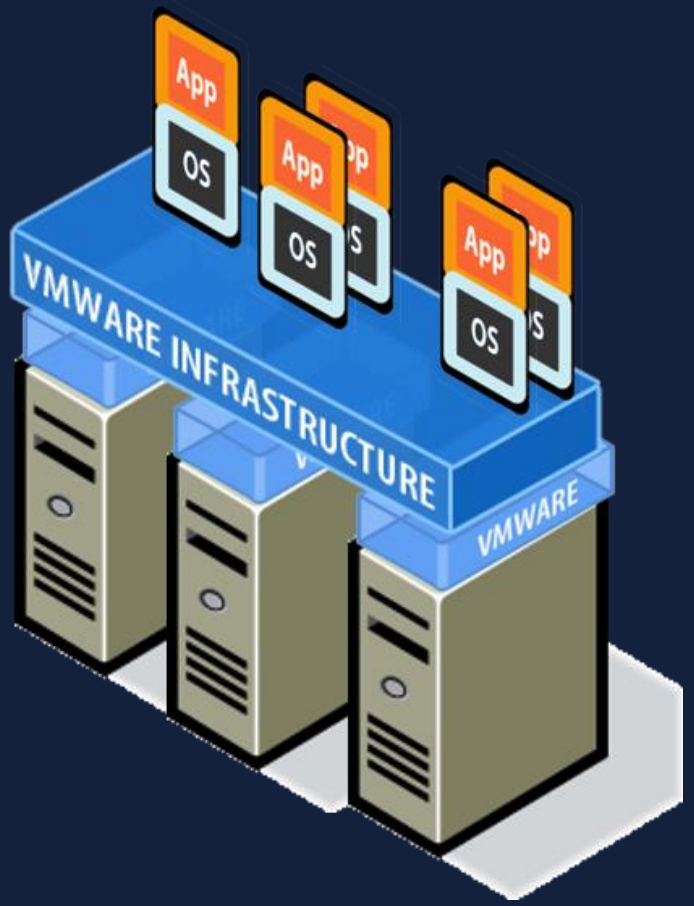
- Problems:
  - VMotion may move VMs across physical ports—network policy must follow
  - Impossible to view or apply policy to locally switched traffic
  - Cannot correlate traffic on physical links—from multiple VMs
- VN-Link:
  - Extends network to the VM
  - Consistent services
  - Coordinated, coherent management



Easy migration from Physical to Virtual

- Consolidation Management with the VMware Infrastructure software will automate the migration from physical to virtual machines.





Easy workflow management for new Virtual Machines

- Provisioning with approval processes with Life Cycle Manager and Stage Manager
- Self Provisioning with Lab Manager



Scalability vServices

- VMware Vmotion, makes it possible to move Virtual Machines, without interrupting the applications running inside.



Automatic Scalability vServices

- VMware Distributed Resource Scheduler automatically balances the Workloads according to set limits and guarantees.
- Removing the need to predict resource assignment.



High Availability vServices

- VMware High Availability makes all Servers and Applications protected against component and complete system failure.
- Only One-Click to configure!



Disaster Recovery vServices

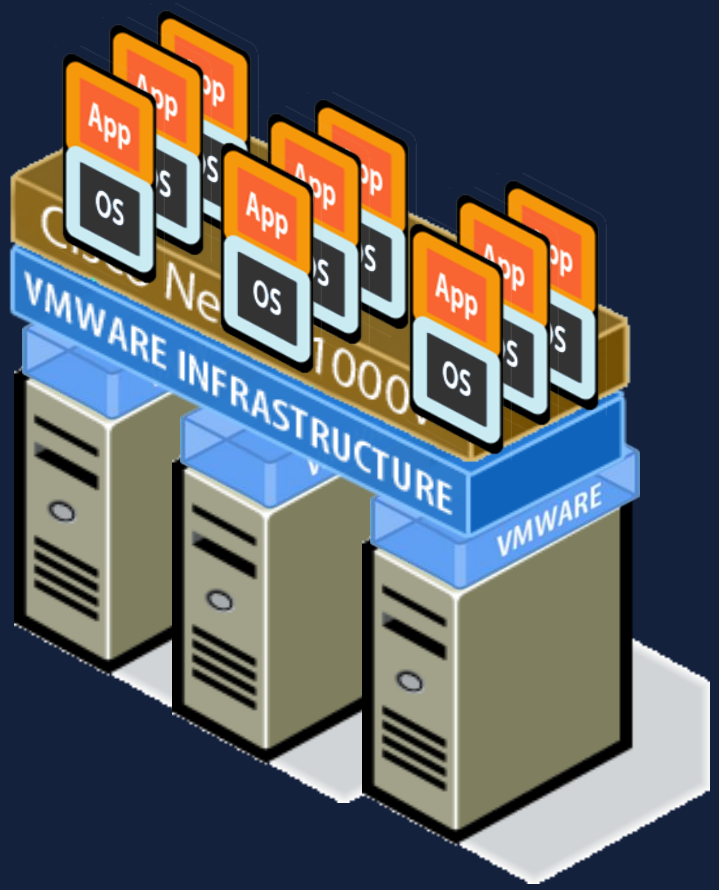
- VMware Site Recovery Manager enables an easy transition from a production site to a Disaster Recovery site.
- Easy Execution for real Disaster
- Easy Testing for good night sleep



compellent 3PAR IBM DELL EQUALLOGIC FalconStor Xiotech EMC NetApp LeftHand NETWORKS HITACHI

Adding Infrastructure vServices

- VMware and Cisco are collaborating to enhance workload mobility and simpler management with virtualization-aware networks
- Nexus 1000V



Security vServices

- The Application vService VMSafe allows security vendors to add superior security solutions inside the VMware Infrastructure.



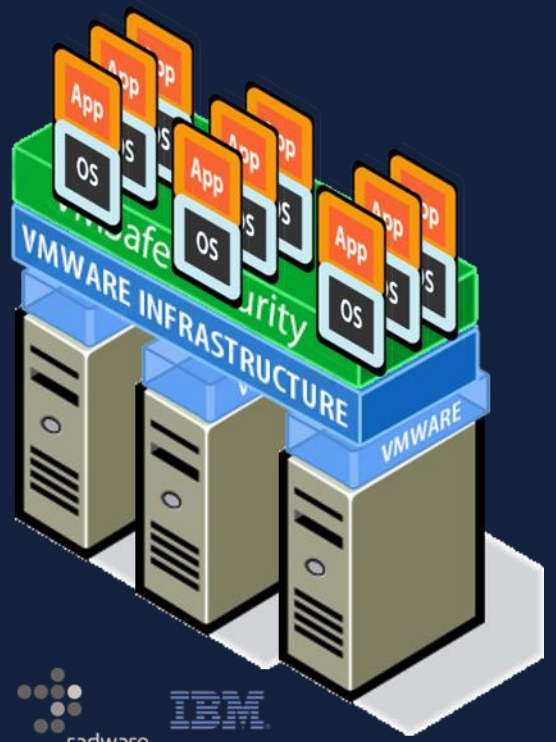
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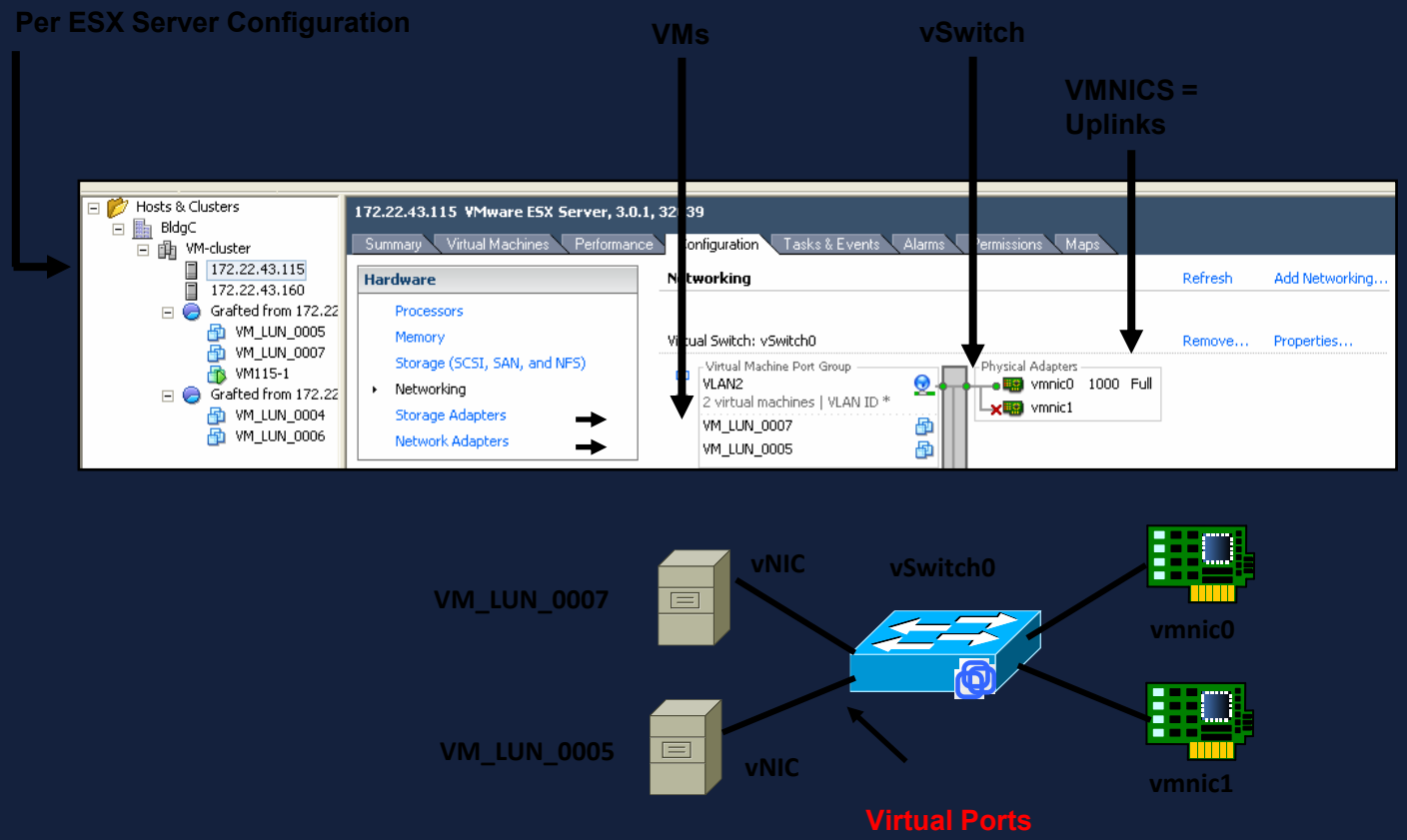
VMware ESX Network Architecture in a Nutshell

- ESX Server Host running Virtual Machines (App + OS) alongside a Console OS
- Physical Hardware: CPU, Memory, network adapters
- Networks: Mgmt Network, VM Kernel Network, Production Network



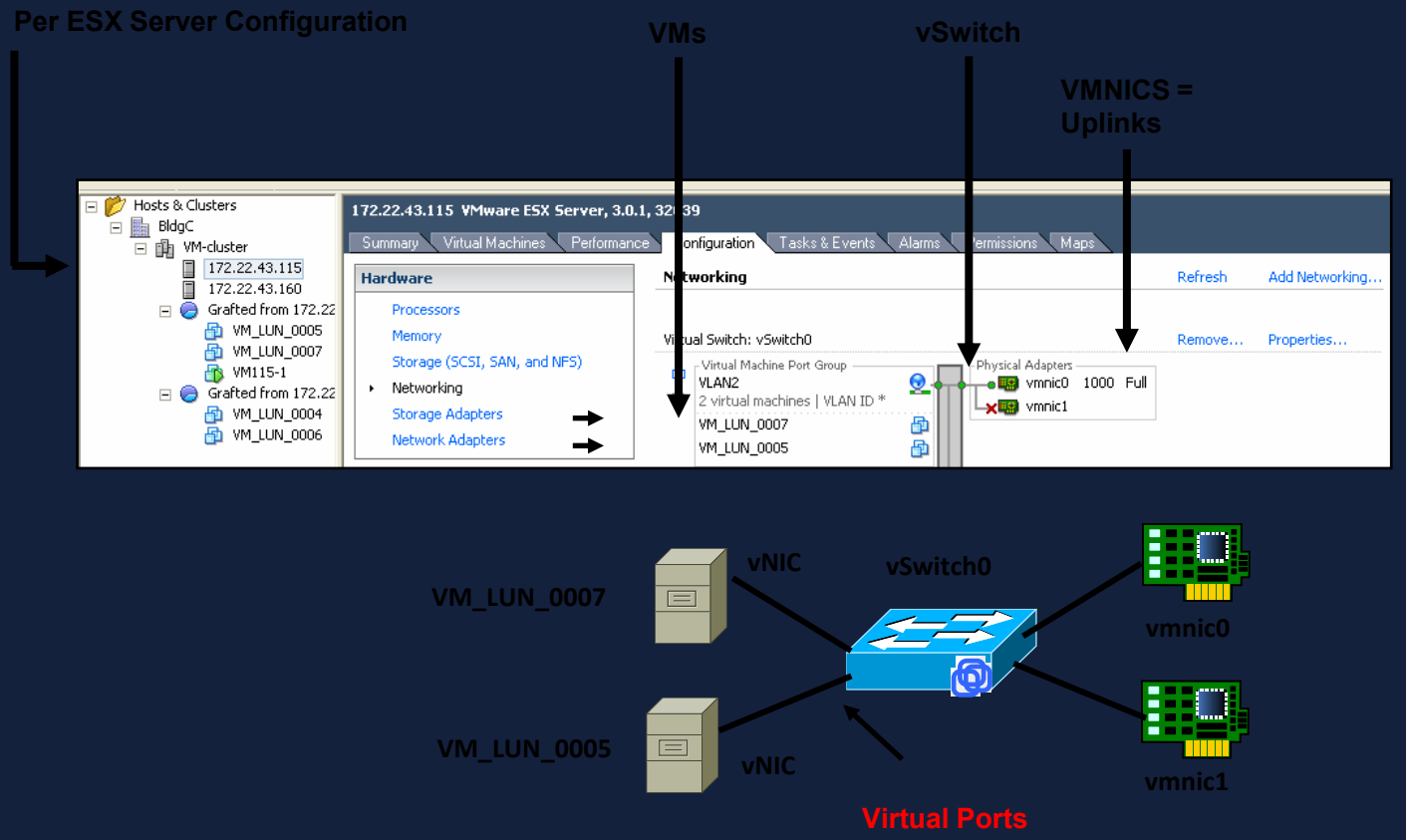
VMware Networking Components

- Per ESX Server Configuration
- VMs connect through vNICs to virtual ports on vSwitch0
- vSwitch
- VMNICS = Uplinks (vmnic0, vmnic1)



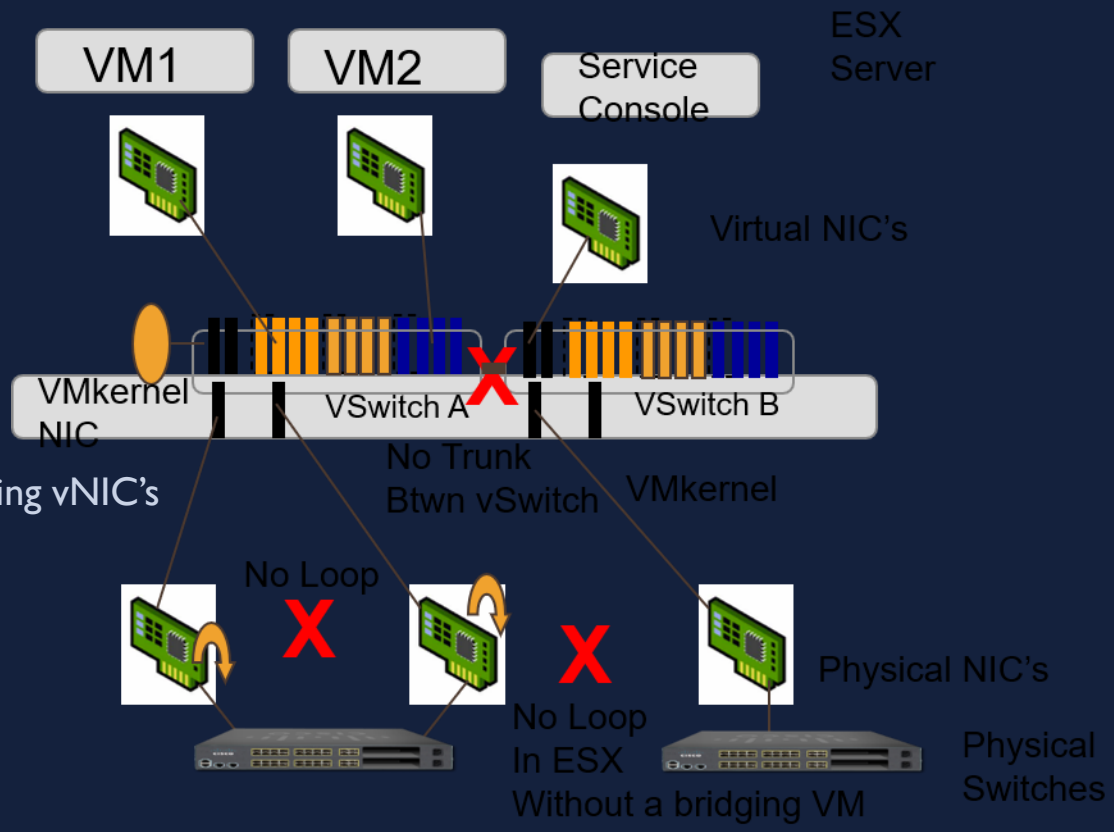
VMware Networking Components (Cont.)

- vSwitch0: Virtual Machine Port Group (Network0) – 10 virtual machines, uplink vmnic0 to the Internet / Trusted LAN VLAN
- vSwitch1: Service Console Port – uplink vmnic1 to the VMotion LAN
- vSwitch2: VMkernel Port – uplink vmnic2



vSwitch Overview

- Software implementation of an Ethernet switch within ESX
- How is it like a switch:
  - MAC addr forwarding
  - VLAN segmentation
- How is it different:
  - No need to learn MAC addresses – it knows the address of the connecting vNIC's
  - No participation in spanning tree
- No trunk between vSwitches; no loop in ESX without a bridging VM



vSwitch Forwarding Characteristics

- Forwarding based on MAC address (no learning): If traffic doesn't match a VM MAC is sent out to vmnic
- VM-to-VM traffic stays local
- Vswitches TAG traffic with 802.1q VLAN ID
- vSwitches are 802.1q-capable
- vSwitches can create EtherChannels

Summary

- Roles of Server Virtualization in an enterprise
- Role of the hypervisor in virtualization.
- Types of hypervisors
- Types of virtualization
- Management techniques for Virtual Servers
- Achieving Virtual Servers scalability
- Networking for Virtual Servers

References

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- http://www.vmware.com/pdf/RVI_performance.pdf
- http://www.vmware.com/pdf/Perf_Best_Practices_vSphere4.1.pdf
- Cisco Server Virtualization with Vmware - http://www.cisco.com/en/US/netsol/ns1148/index.html